Architecture Decision Records

Why the system is built the way it is

Short records of the main design decisions, so the team and judges can see why each was made.

ID	Decision	Why	Rejected alternative
ADR-01	Baselines per role, not per agent	New and short-lived agents have no history; a role baseline covers them on their first event.	Per-agent baselines (cold-start problem)
ADR-02	Rules inline, ML asynchronous	The gateway must add only milliseconds; heavy scoring cannot sit in the request path.	Everything inline (latency); everything async (cannot block in time)
ADR-03	Taint tracking for prompt injection	Content cannot be reliably classified as malicious; tracking what influenced an action can be checked.	Scanning documents for injection text (easy to evade)
ADR-04	Markov sequence model plus CUSUM	Cheap, explainable, works on little data; CUSUM catches slow drift under a per-event threshold.	Transformer sequence model (data hungry, hard to explain)
ADR-05	Weighted score with hard overrides	Transparent weights give clear reasons; overrides make honeypot and kill-chain cases certain.	Learned black-box score (cannot explain alerts)
ADR-06	Signed, versioned manifests	Separates approved change from drift and prevents silent edits to what 'normal' means.	Auto-learning the role definition (an attacker could teach it)
ADR-07	Fail closed for sensitive, open for low-risk	A detector outage must not open sensitive data, nor stop all legitimate work.	Always fail closed (outage stops business); always fail open (attack window)
ADR-08	Honeypots and decoys	No legitimate agent touches them, so access is near-certain evidence even if the attacker copies normal behaviour.	Relying on statistics alone (evadable)
ADR-09	Shadow mode for new rules and models	Lets us measure false positives on live traffic before anything can block work.	Enable immediately (risk of disruption)

Template for new decisions

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ADR-NN: <decision in one line>
Status: proposed | accepted | superseded by ADR-MM
Context: what forced the decision
Decision: what we chose
Consequences: what gets easier, what gets harder
Rejected: alternatives and why
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